QUIZ - 1 (Not in the syllabus)

You are given a matrix **A**. Write a Java program that will **create and print a new Matrix M** where:

$$M = 2*A^T$$

[Here, T means transpose. Your code should work for any given matrix of any size.]

Given Array	Output

`int [][] A = {{3,4},`
`{6,7},`
`{2,2}};`
	6 12 4
	8 14 4

`int [][] A = {{13,24,13},` `{22,17,11}};`	26 44 48 34 26 22

QUIZ 2:

Design the **SlackChannel** class in such a way so that the following code provides the expected output. **[Marks: 13]**

Driver Class	Output

`public class SlackTester{`
`  public static void main(String args[]){`
`  SlackChannel c1 = new SlackChannel("CSE111 Summer 2025");`
`  System.out.println("1--------------");`
`  c1.addMember("KNI", "Faculty");`
`  c1.addMember("PRN", "Faculty");`
`  System.out.println("2--------------");`
`  c1.info();`
`  System.out.println("3--------------");`
`  c1.addMember("T14", "ST");`
`  c1.addMember("Aftab");`
`  c1.addMember("Mou");`
`  c1.addMember("T65", "ST");`
`  System.out.println("4--------------");`
`  c1.info();`
`  System.out.println("5--------------");`
`  SlackChannel c2 = new SlackChannel("CSE110 Summer 2025");`
`  System.out.println("6--------------");`
`  c2.addMember("KNI", "Faculty");`
	A new channel has been created
	1--------------
	Faculty KNI has been added to slack
	Faculty PRN has been added to slack
	2--------------
	Channel Name: CSE111 Summer 2025
	Total Members: 2
	Students:
	Other Members:
	KNI
	PRN
	3--------------
	ST T14 has been added to slack
	Student Aftab has been added to slack
	Student Mou has been added to slack
	Maximum capacity exceed
	4--------------
	Channel Name: CSE111 Summer 2025
	Total Members: 5
	Students:
	Aftab
	Mou
	Other Members:
	KNI
	PRN

c2.addMember("Ria");
c2.addMember("T45", "ST");
System.out.println("7--------------");
c2.info();
}
}
	T14
	5--------------
	A new channel has been created
	6--------------
	Faculty KNI has been added to slack
	Student Ria has been added to slack
	ST T45 has been added to slack
	7--------------
	Channel Name: CSE110 Summer 2025
	Total Members: 3
	Students:
	Ria
	Other Members:
	KNI
	T45

2. Trace the table below and write down the outputs in the question paper. Your answers won't be accepted without workings. **[Marks: 7]**

1	public class A {
2	public int temp = 4;
3	public int sum, y, x;
4	public void methodA(int m){
5	int [] n = {2,5};
6	int x = 0;
7	y = m + this.methodB(x++,m)+(temp++);
8	x = this.x + 2 + n[0];
9	sum = sum + x + y;
10	n[0] = sum + 2;
11	System.out.println(n[0]+" " + x+ " " + sum);
12	}
13	public int methodB(int m, int n){
14	int y = 4 + this.y + m;
15	x = this.y + y + (++temp) - n;
16	sum = x + y + this.sum;
17	System.out.println(y+ " " + this.x + " " +sum);
18	return x;
19	}
20	}

```
public class Tester {
   public static void main(String [] args){
       A t1 = new A();
       t1.methodA(5);
       A t2 = new A();
       t2.methodB(12, 2);
   }
}
```

Outputs		
	4	
		35

QUIZ - 3 (After midterm)

Design a **BookshopOrder** class to generate the following output for the given driver code.

Driver Code	Output

public class BookshopOrderTester {
  public static void main(String[] args) {
  BookshopOrder.BookshopInfo();
  System.out.println("=====1=====");
  BookshopOrder order1 = new BookshopOrder("The Great Gatsby", 300.0);
  BookshopOrder order2 = new BookshopOrder("Data Structures", 400.0);
  order1.displayInfo();
  order2.displayInfo();
  System.out.println("=====2=====");
  BookshopOrder.BookshopInfo();
  System.out.println("=====3=====");
  order1.completeOrder();
  System.out.println("=====4=====");
  order1.displayInfo();
  System.out.println("=====5=====");
  BookshopOrder.BookshopInfo();
  BookshopOrder order3 = new BookshopOrder("Physics Fundamentals", 500.0);
  System.out.println("=====6=====");
  order3.completeOrder();
  System.out.println("=====7=====");
  order3.displayInfo();
  System.out.println("=====8=====");
  BookshopOrder order4 = new BookshopOrder("History of Art", 350.0);
  order4.completeOrder();
  System.out.println("=====9=====");
  BookshopOrder.BookshopInfo();
  System.out.println("=====10=====");
	Bookshop Status:
	Active Orders: 0
	Total Revenue: Tk0.0
	=====1=====
	Order Details:
	Book Title: The Great Gatsby
	Order Amount: Tk300.0
	Status: active
	Order Details:
	Book Title: Data Structures
	Order Amount: Tk400.0
	Status: active
	=====2=====
	Bookshop Status:
	Active Orders: 2
	Total Revenue: Tk0.0
	=====3=====
	Order completed. Revenue added: Tk300.0
	=====4=====
	Order Details:
	Book Title: The Great Gatsby
	Order Amount: Tk300.0
	Status: completed
	=====5=====
	Bookshop Status:
	Active Orders: 1
	Total Revenue: Tk300.0
	=====6=====
	Order completed. Revenue added: Tk500.0
	=====7=====
	Order Details:

BookshopOrder.payTax(10); } }	Book Title: Physics Fundamentals Order Amount: Tk500.0 Status: completed =====8===== Order completed. Revenue added: Tk350.0 =====9===== Bookshop Status: Active Orders: 1 Total Revenue: Tk1150.0 =====10===== Tax paid: Tk115.0

QUIZ - 4

Carefully read the following School and Tester classes and write the child class "**Department**" so that the following output is produced.

Driver Code	**Output**

public class School {
  public static int total_rooms = 100;
  protected String name;
  double budget;
  public School(String name, double budget) {
  this.name = name;
  this.budget = budget;
  }
  public String toString() {
  return "Name: "+ name +", School Budget: "+this.budget + " taka\nRooms left: "+ School.total_rooms + "\n";
  }
}
 	
public class Tester {
 public static void main(String[] args) {
  Department cse = new Department("CSE", 2000, "AI Research");
  Department mns = new Department("MNS", 4500, "Renewable Energy");
  System.out.println("========1========");
  cse.allocateRooms(30);
  System.out.println("========2========");
  System.out.println(cse);
  System.out.println("========3========");
  mns.allocateRooms(40);
  System.out.println("========4========");
  System.out.println(mns);
  System.out.println("========5========");
  cse.addProjects("Cybersecurity");
  cse.addProjects("NLP");
	========1========
	30 rooms have been allocated to CSE
	========2========
	Name: CSE, School Budget: 2000.0 taka
	Rooms left: 70
	Allocated Rooms: 30, Department Fund: 0.0 taka
	Projects:
	AI Research
	========3========
	40 rooms have been allocated to MNS
	========4========
	Name: MNS, School Budget: 4500.0 taka
	Rooms left: 30
	Allocated Rooms: 40, Department Fund: 0.0 taka
	Projects:
	Renewable Energy
	========5========
	========6========
	Name: CSE, School Budget: 2000.0 taka
	Rooms left: 30
	Allocated Rooms: 30, Department Fund: 0.0 taka
	Projects:
	AI Research, Cybersecurity, NLP
	========7========
	Could not add LLM
	========8========
	Cannot request more than 50% of total budget for CSE.
	========9========

```
   System.out.println("========6========");
   System.out.println(cse);
   System.out.println("========7========");
   cse.addProjects("LLM");
   System.out.println("========8========");
   cse.requestFund(1001);
   System.out.println("========9========");
   mns.requestFund(3300);
   System.out.println("========10========");
   cse.requestFund(1000);
   System.out.println("========11========");
   cse.allocateRooms(31);
   System.out.println("========12========");
   System.out.println(cse);
   System.out.println("========13========");
   System.out.println(mns);
 }
}
```

```
========10========
Not enough rooms.
========11========
Name: CSE, School Budget: 1000.0 taka
Rooms left: 30
Allocated Rooms: 30, Department Fund: 1000.0
taka
Projects:
AI Research, Cybersecurity, NLP
========12========
Name: MNS, School Budget: 1200.0 taka
Rooms left: 30
Allocated Rooms: 40, Department Fund: 3300.0
taka
Projects:
Renewable Energy
```